

Tech & Teach

Digital Chip Design Program

Design RTL

names:

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Task :11

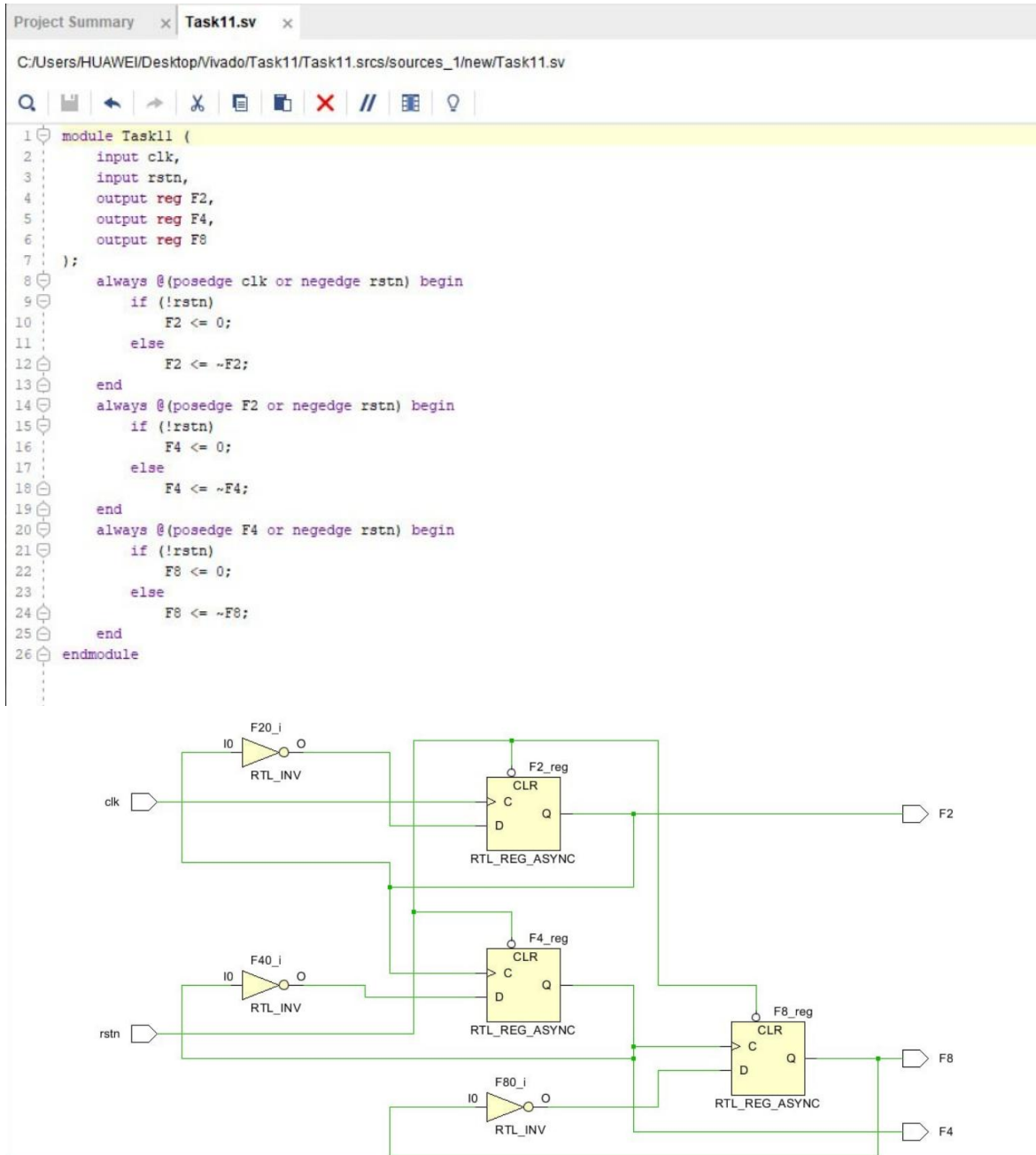
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Task topic:

Seq & Comb Circuits Datapath – Part 2

Version 1.0

Task: 1.1

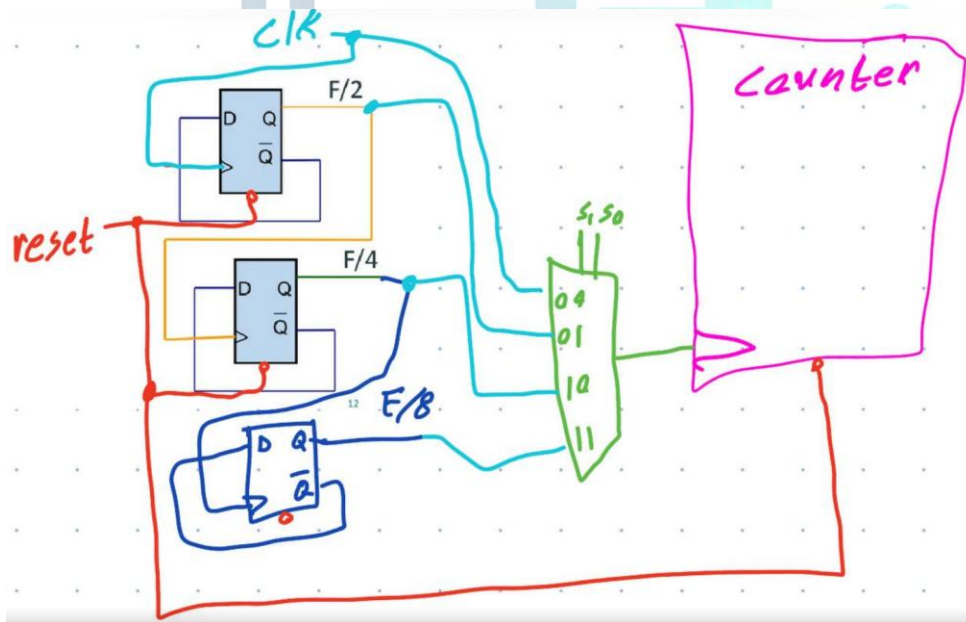


Task: 1.5

1.1.5

We need three D-FF for reducing frequency
Also we need Mux to choose which frequency we will use
At the end we need counter to count

Task: 1.6



1.7

```

1 module counter #(parameter n = 4) (
2     input logic clk, rstn,
3     output logic [n-1:0] Q
4 );
5     always@(posedge clk, negedge rstn)
6         if(!rstn)
7             Q <= 0;
8         else
9             Q <= Q+1;
10 endmodule

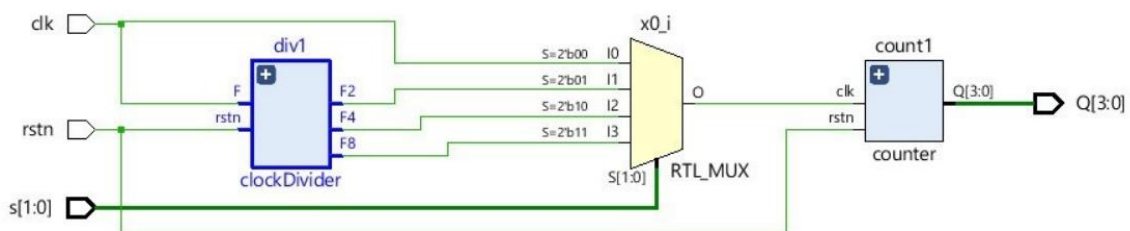
11 module clockDivider (F, rstn, F2, F4, F8);
12     input logic F, rstn;
13     output logic F2, F4, F8;
14     always @(posedge F, negedge rstn)
15         if(rstn) begin
16             F2 <= ~F2;
17         end
18         else F2 = 0;
19     always @(posedge F2, negedge rstn)
20         if(rstn) begin
21             F4 <= ~F4;
22         end
23         else F4 = 0;
24     always @(posedge F4, negedge rstn)
25         if(rstn) begin
26             F8 <= ~F8;
27         end
28         else F8 = 0;
29 endmodule

```

```

1 module top#(parameter n = 4) (
2     input logic clk, rstn,
3     input logic [1:0] s ,
4     output logic [n-1:0] Q);
5     logic [2:0] w;
6     logic x0;
7     clockDivider div1(
8         .F(clk),
9         .rstn(rstn),
10        .F2(w[0]),
11        .F4(w[1]),
12        .F8(w[2])
13    );
14    always_comb begin
15        case(s)
16            2'b00 : x0 = clk;
17            2'b01 : x0 = w[0];
18            2'b10 : x0 = w[1];
19            2'b11 : x0 = w[2];
20            default : x0 = clk;
21        endcase
22    end
23    counter #(n) count1(
24        .clk(x0),
25        .rstn(rstn),
26        .Q(Q)
27    );
28 endmodule
29

```



1.8



```
1  `timescale 1ns/1ps
2  module top;
3
4      parameter N = 4;
5      logic      clk;
6      logic      rstn;
7      logic [1:0] sel;
8      logic [N-1:0] Q;
9      speed_controlled_counter #(
10         .N(N)
11     ) dut (
12         .clk(clk),
13         .rstn(rstn),
14         .sel(sel),
15         .Q(Q)
16     );
17
18     always #5 clk = ~clk;
19     initial begin
20         clk = 0;
21         rstn = 0;
22         sel = 2'b00;
23         #20;
24         rstn = 1;
25         sel = 2'b00;
26         #100;
27         sel = 2'b01;
28         #200;
29         sel = 2'b10;
30         #400;
31         sel = 2'b11;
32         #800;
```



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Work distribution

Section	shaher	Suliman	Abdullah
Report	✓		
Task 1.1, 1.2			✓
Task 1.5,1.6		✓	
Task 1.7		✓	

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